State	λ	+	0	1
0 initial	-	-	1 | 0 | R	1 | 1 | R
1	-	2 | + | L	1 | 0 | R	1 | 1 | R
2	-	-	3 | 0 | D	3 | 1 | D
3	4 | + | U	-	-	-
4	15 | 0 | R	-	5 | 0 | R	5 | 1 | R
5	-	6 | + | R	5 | 0 | R	5 | 1 | R
6	7 | λ | L	-	6 | 0 | R	6 | 1 | R
7	-	16 | λ | D	12 | λ | D	8 | λ | D
8	9 | λ | L	-	-	-
9	9 | λ | L	10 | + | U	-	-
10	-	-	11 | 1 | D	18 | 0 | L
11	-	3 | λ | L	-	-
12	13 | λ | L	-	-	-
13	13 | λ | L	14 | + | U	-	-
14	-	-	11 | 0 | D	11 | 1 | D
15	-	-	5 | 0 | R	5 | 1 | R
16	16 | λ | L	17 | λ | U	-	-
17	-	-	21 | 0 | S	21 | 1 | S
18	20 | 1 | D	-	19 | 1 | D	18 | 0 | L
19	20 | λ | R	-	-	-
20	20 | λ | R	3 | λ | L	-	-
21 halt	21 | λ | S	21 | + | S	21 | 0 | S	21 | 1 | S

Entry: next state | written symbol | motion

λ = blank - = unobserved S = stop